

Written Assignment 4- Semantics Analysis & TAC Generation

1. Please construct an annotated parse tree for the input string $4*5+6$

where the syntax-directed definition is as following

Productions	Semantic Rules
$E \rightarrow E1 * T$	$E.val = E1.val * T.val$
$E \rightarrow T$	$E.val = T.val$
$T \rightarrow T1 + F$	$T.val = T1.val + F.val$
$T \rightarrow F$	$T.val = F.val$
$F \rightarrow i$	$F.val = i.lexval$

2. Please construct an annotated parse tree for the input string

$4+(5*6+9)*7$ where the syntax-directed definition is as following

Productions	Semantic Rules
$E \rightarrow E1 * T$	$E.val = E1.val * T.val$
$E \rightarrow T$	$E.val = T.val$
$T \rightarrow T1 + F$	$T.val = T1.val + F.val$
$T \rightarrow F$	$T.val = F.val$
$F \rightarrow (E)$	$F.val = E.val$
$F \rightarrow i$	$F.val = i.lexval$

3. For grammar $G(S)$, Please design a translation scheme that prints the position of each a in a sentence.

$$\begin{aligned} S' &\rightarrow S \\ S &\rightarrow (L) | a \\ L &\rightarrow L, S | S \end{aligned}$$

For example, when the sentence is

$$(a, (a, (a, a), (a)),$$

the printed result is

$$2 \ 5 \ 8 \ 10 \ 14.$$

4. Please construct a bottom-up parsing tree and its corresponding dependency graph for a declaration

$$\text{int } x, y, z, r;$$

using the following SDD.

PRODUCTION	SEMANTIC RULES
1) $D \rightarrow T L$	$L.inh = T.type$
2) $T \rightarrow \text{int}$	$T.type = \text{integer}$
3) $T \rightarrow \text{float}$	$T.type = \text{float}$
4) $L \rightarrow L_1, \text{id}$	$L_1.inh = L.inh$ $addType(id.entry, L.inh)$
5) $L \rightarrow \text{id}$	$addType(id.entry, L.inh)$

Figure 5.8: Syntax-directed definition for simple type declarations

5. Please translate the following program fragment into three-address code using the form of short circuit code.

```
i=2;
loop=0;
while (loop==0 && i<=10) {
    j=1;
    while (loop ==0 && j<i)
        if (a[i,j] == x)
            loop=1;
        else j=j+1;
    if (loop==0) i=i+1;
}
```

Notes: Here we assume that the declaration of array a is array [0..10,0..10], each data element of array a would **use 2 storage units**, and the start address of array A's storage area is a.

6. Translate the following program fragment into three-address code of quadruple representation, using short circuit code and back-patching techniques. Please keep the traces of merging and backpatching.

```
i=2;
m=0;
loop=0;
while (loop==0 && i<=10) {
    j=1;
    while (loop ==0 && j<=i)
        if (a[i,j] != a[j,i]) //"!=" means "not equal to"
        {
            loop=1;
            m=1;
        }
```

```
    }  
    else j=j+1;  
    if (loop==0) i=i+1;  
}
```

Notes: Here we assume that the declaration of array A is array [1..10,1..10], each data element of array a would **only use 1 storage unit**, and the start address of array a's storage area is a.

Complete the homework and submit it before the next next class to the teaching assistant (Room 156, Building of Sch. Computer Sci&Eng).

1. Everyone should give their completed workbook to the teaching assistant before the designated time. There is a box at the door of the room specifically for collecting homework.
2. After the first submission of each assignment, it will be graded based on the quality of completion. Allow the next submission to modify errors from the previous assignment. The final score for each assignment will be given based on the final quality and number of revisions made.